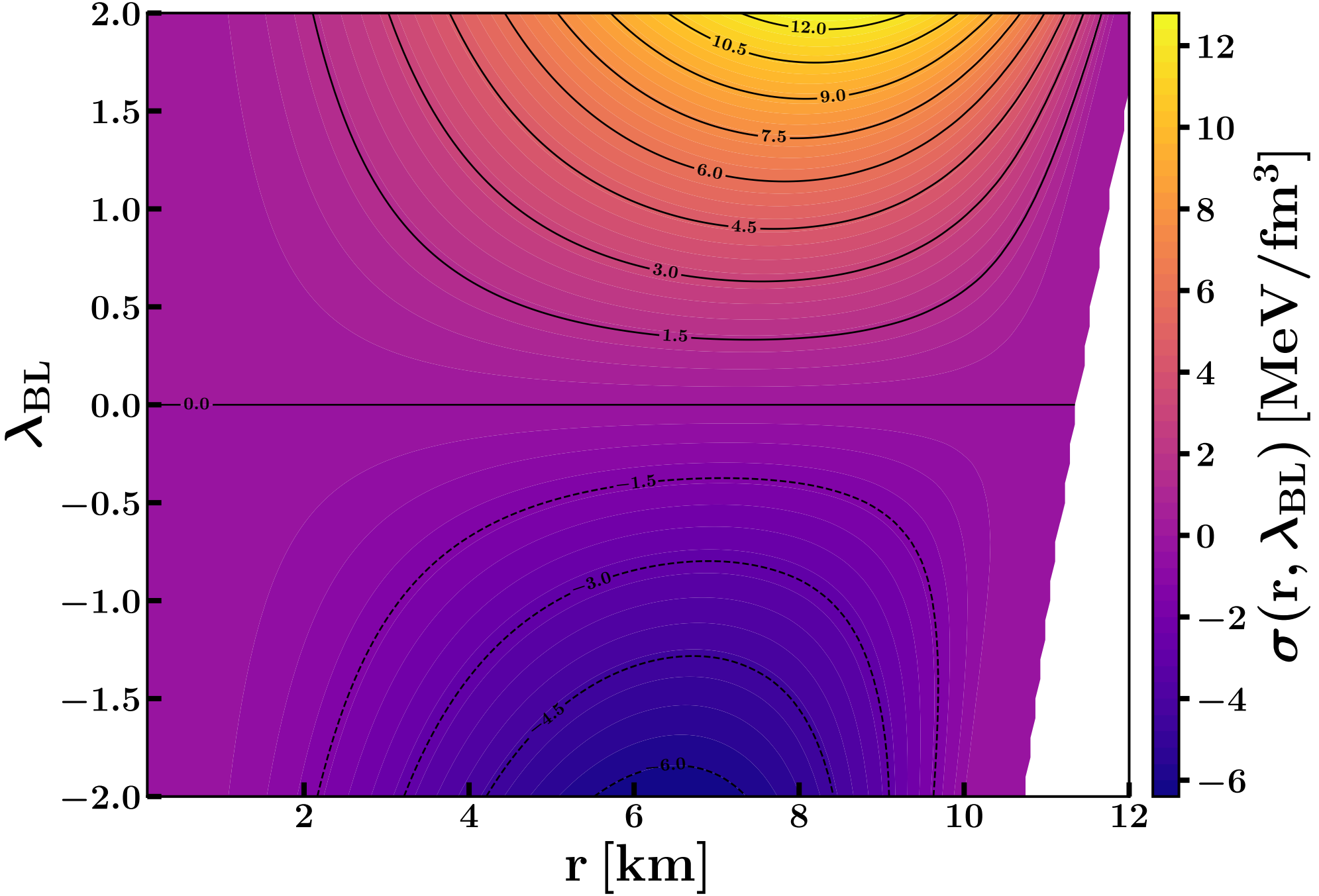
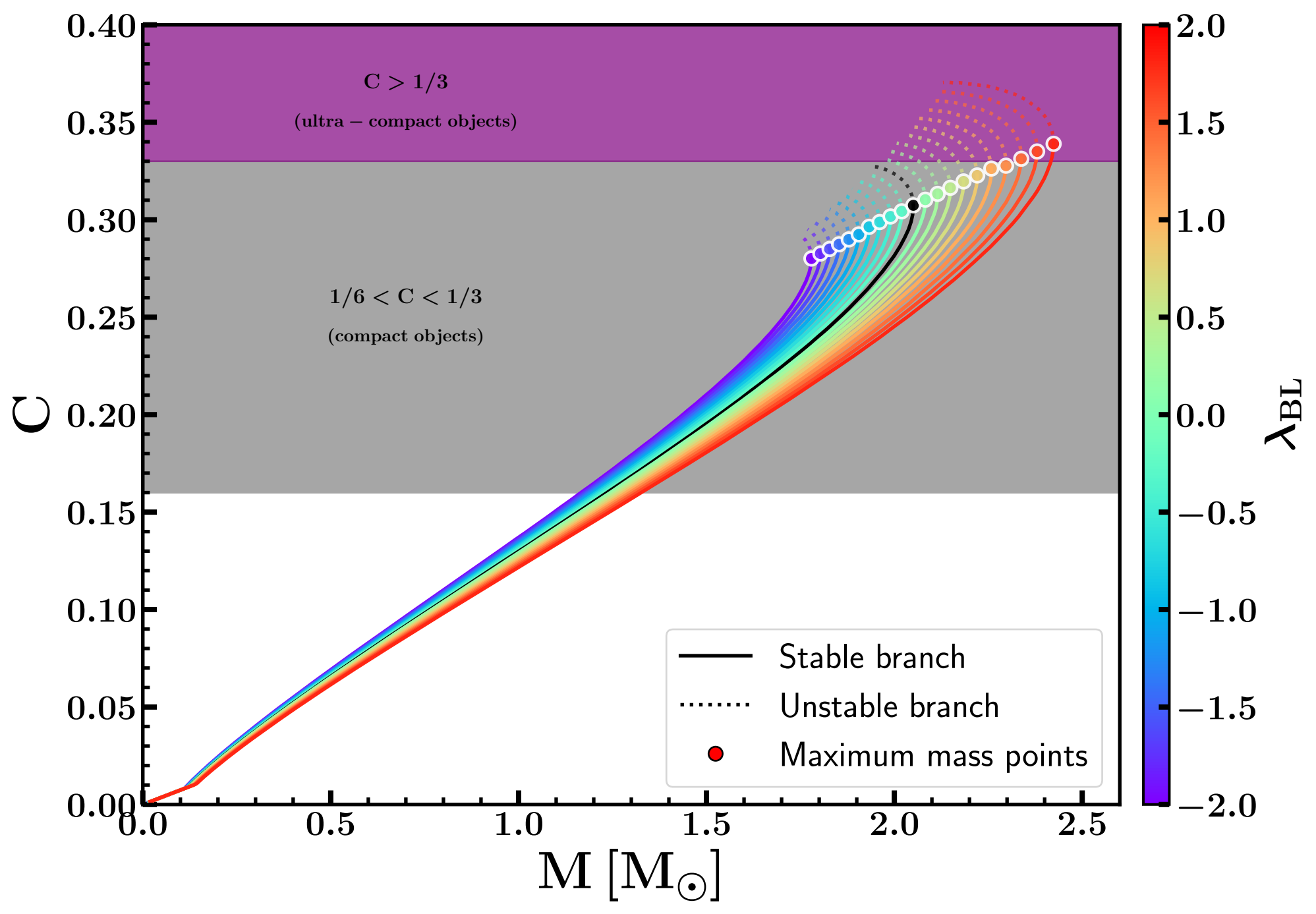
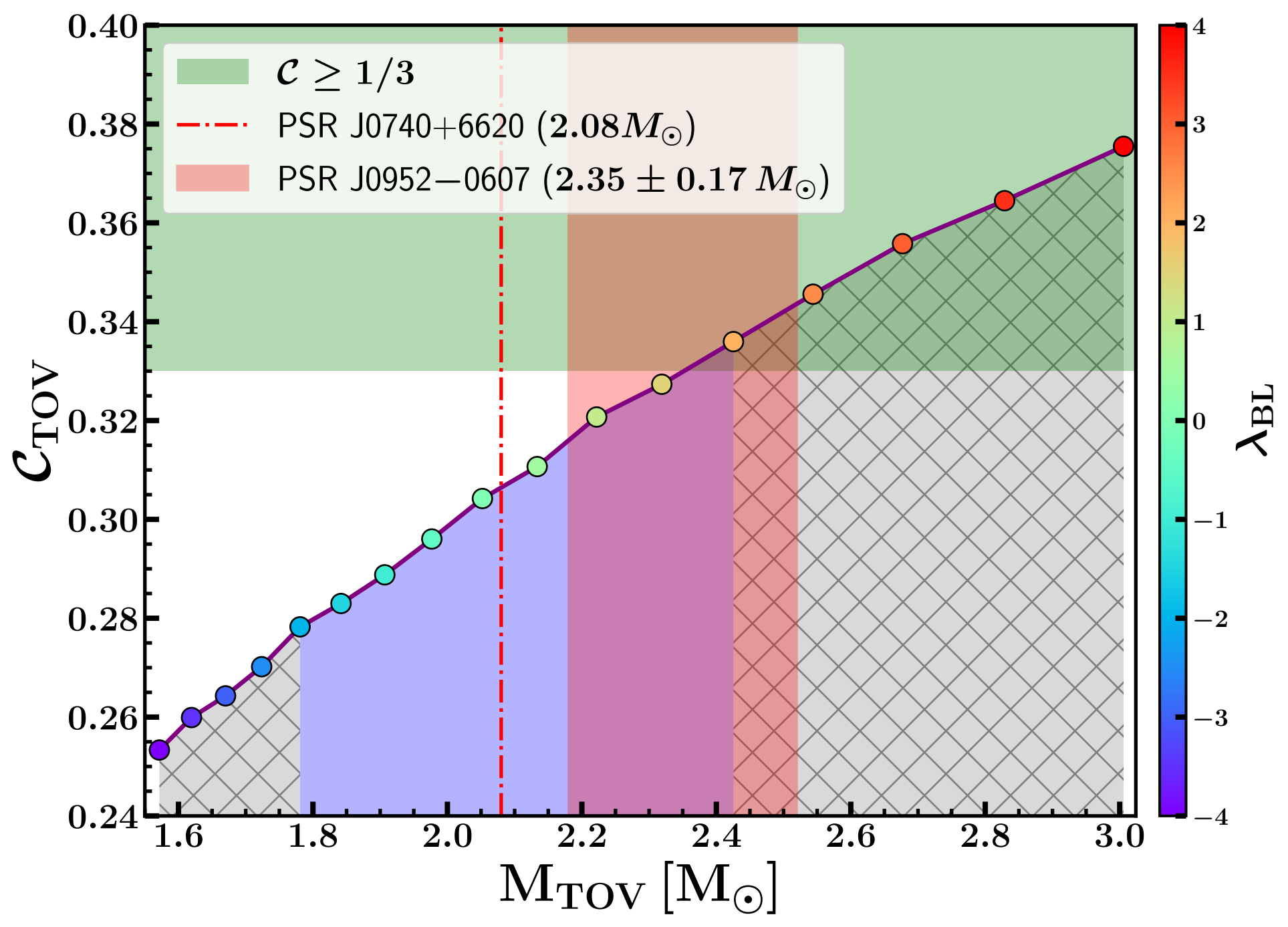
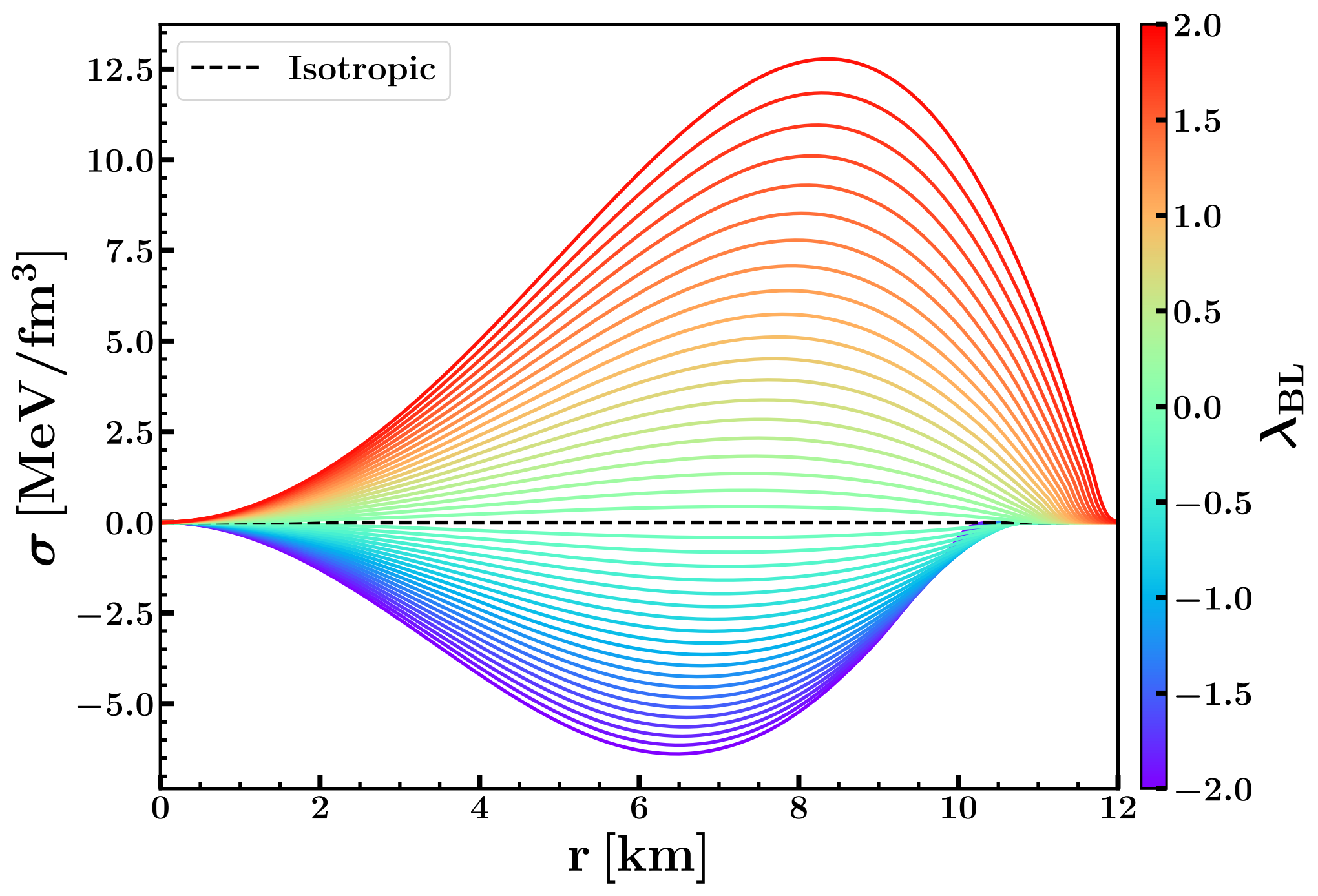
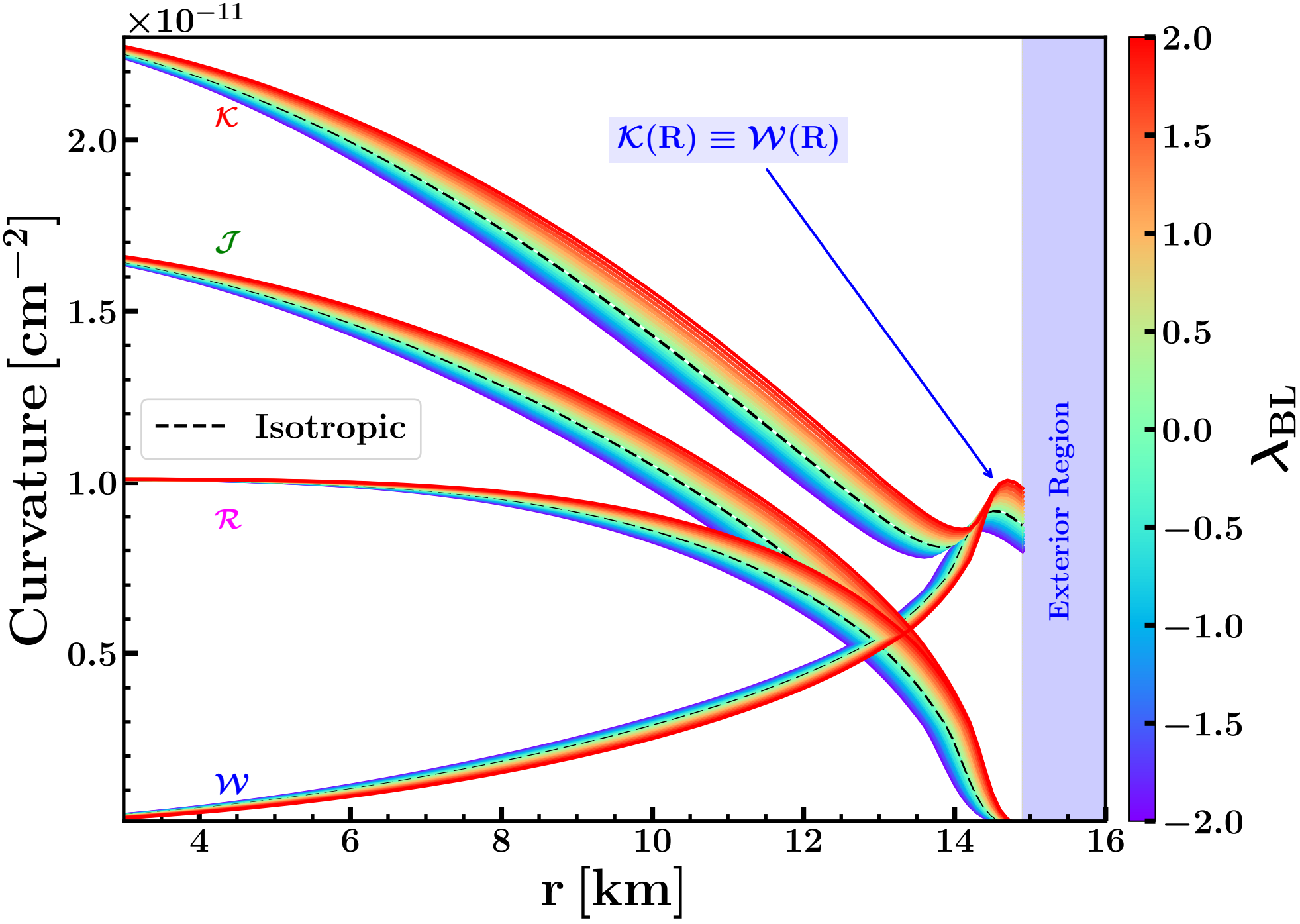
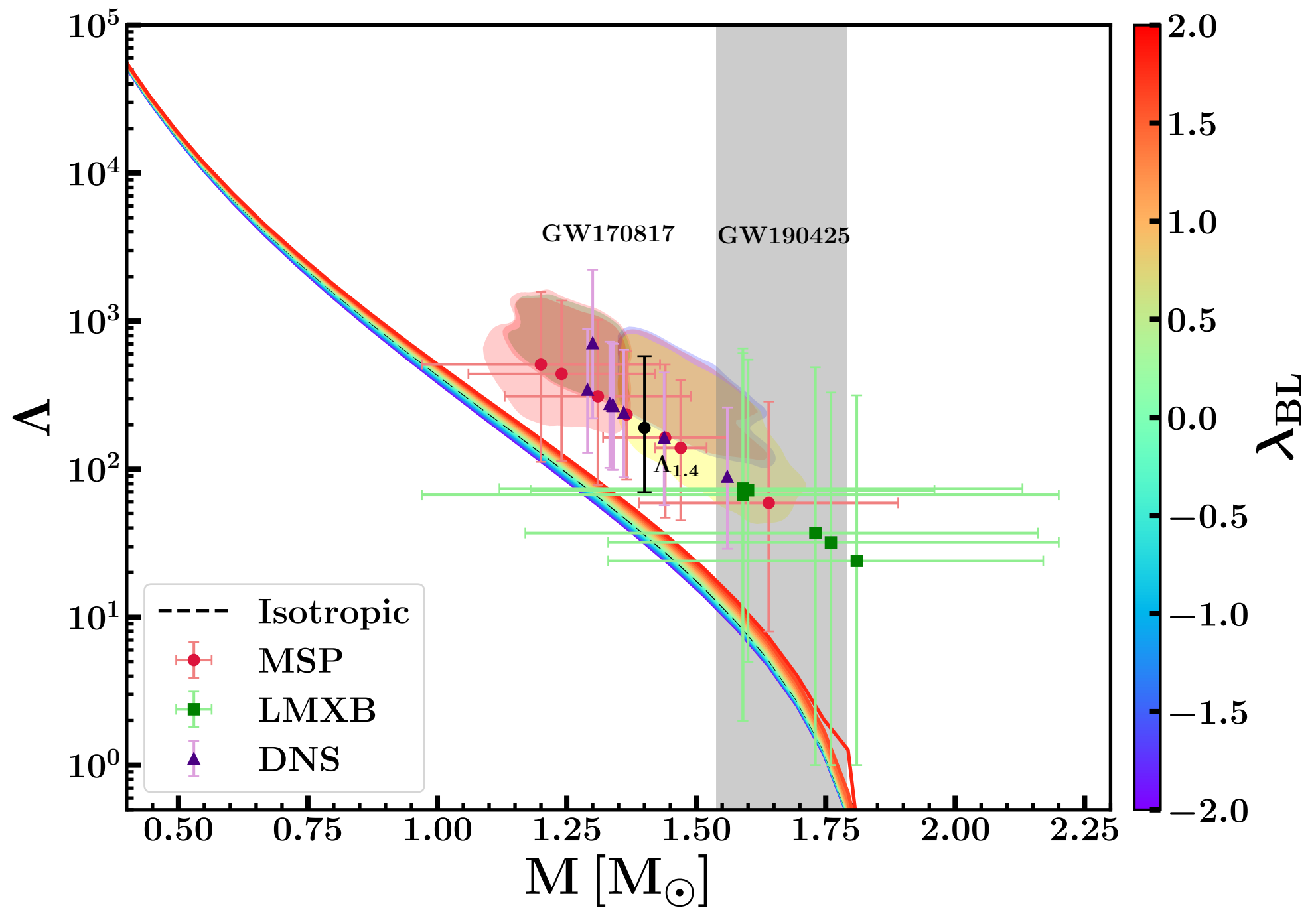

12.0
10.5
9.0
7.5
6.0
4.5
3.0
1.5
0.0
1.5
3.0
4.5
6.0
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
λ_{BL}
2
4
6
8
10
12
r [km]
12
10
8
6
4
2
0
−2
−4
−6
$\sigma(\mathrm{r}, \lambda_{\mathrm{BL}})$ [MeV/fm^3]


C > 1/3
(ultra − compact objects)
1/6 < C < 1/3
(compact objects)
Stable branch
Unstable branch
Maximum mass points
C
M [M⊙]
λBL
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.0
0.5
1.0
1.5
2.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0


$\mathcal{C} \geq 1/3$
PSR J0740+6620 ($2.08 M_\odot$)
PSR J0952−0607 ($2.35 \pm 0.17\, M_\odot$)
$\mathcal{C}_{\rm TOV}$
$M_{\rm TOV}$ [$M_\odot$]
$\lambda_{\rm BL}$
0.40
0.38
0.36
0.34
0.32
0.30
0.28
0.26
0.24
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
4
3
2
1
0
−1
−2
−3
−4


Isotropic
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
σ [MeV/fm³]
0
2
4
6
8
10
12
r [km]
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
λBL


$\times 10^{-11}$
$\mathcal{K}$
$\mathcal{J}$
$\mathcal{R}$
$\mathcal{W}$
$\mathcal{K}(\mathrm{R}) \equiv \mathcal{W}(\mathrm{R})$
Isotropic
Exterior Region
Curvature [cm^{-2}]
r [km]
λ_{BL}
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
4
6
8
10
12
14
16


10^5
10^4
10^3
10^2
10^1
10^0
Λ
GW170817
GW190425
$\Lambda_{1.4}$
Isotropic
MSP
LMXB
DNS
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
M [$M_\odot$]
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
λ_{BL}